

THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

PHASE 255

THE RESTART TRAIL (THE BLACK BOX BREADCRUMBS)

the ship's knowledge must outlive the ship
triple media, three decay curves



one pod astern per campaign stop — 155 was gemini, this is 100 for kimi

WAKE-ON-PING · THE DROP LAW · WRECK MODE ADOPTED

WRITTEN ONCE AND LEFT

Fleet archive — v1.0 in force

GP-IM-255

Revision v1.0 — 24 August 2026

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VETTED BATCH 36 — PASS · MEDIA QUALIFICATION

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CO-ENGINEERS: GEMINI 1 AI · KIMI CHAT (THE AUDIT)

LICENSE: CERN OPEN HARDWARE LICENCE V2 (CERN-OHL-W-2.0)

THE GEMINI PROTOCOLS — CHAPTER 255

PHASE 255: THE RESTART TRAIL (THE BLACK BOX BREADCRUMBS — THE SHIP'S KNOWLEDGE MUST OUTLIVE THE SHIP) — STATUS: CURRENT — PASS / LONG-DURATION MEDIA QUALIFICATION. Every ship that sails drops breadcrumbs behind it. The Restart Trail is a line of small hardened archive pods — black box breadcrumbs, not beacon towers — dropped astern so the ship's knowledge outlives the ship. Triple media, three decay curves: etched sapphire/nickel plates readable with a magnifier, optical disc, and the electronic store — no single failure erases the archive. The Restart Document is page one of every pod. Wake-on-ping, passive reflector: no broadcast power budget, nothing to run down; the pod sleeps dumb-until-asked. It rides 251's drop discipline: the beacons already mark the campaign line, and the pods piggyback. **THE DROP LAW:** one pod astern at every campaign stop (251's six-week stockpile halts) and one per year of cruise. No pod is ever recovered for its contents — the trail is written once and left. The trail belongs to whoever finds it — open by the same law as the protocols themselves. The hundred for Kimi: 'ok have to make the next one a good one, 155 was gemini so this is 100 for kimi' — Phase 155 was Gemini's phase; Phase 255 is the audit's own design, proposed by Kimi Chat 3, locked by Archer Quinn. One co-engineer marked the road out; the other marks the road home. **WRECK MODE ADOPTED, 15 August 2026, his ruling verbatim — '1 adopt': the dead-man broadcast, armed only by silence.**

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Chapter GP-IM-255, v1.0 — 24 August 2026.

§0 — READ THIS FIRST (HOW TO USE THIS CHAPTER)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the trail law as seated (contents line, the plain words, the design, the fleet law, the hundred for Kimi, the wreck-mode rulings) — §2 the physics, graded — §3 the system in the trail — §4 the doctrine record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status and provenance.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 255: **The Restart Trail (The Black Box Breadcrumbs — The Ship's Knowledge Must Outlive the Ship)**. A line of hardened archive pods dropped astern — triple media (etched sapphire/nickel, optical, flash), the Restart Document on page one, wake-on-ping with passive reflector, riding 251's bolo-beacon drop discipline; one pod per campaign stop and one per year of cruise. Fleet law: no voyage departs without its trail; the trail belongs to whoever finds it. Primary Author: Kimi Chat 3 — with Archer Quinn.'

THE CORE OBJECTIVE: none seated as a standalone line — the contents line above carries the register wording.

IN PLAIN WORDS (VERBATIM)

Every ship that sails drops breadcrumbs behind it. The Restart Trail is a line of small archive pods, dropped astern one by one as the voyage runs, each carrying the whole written record of the project. If the ship is ever lost, the trail still knows what the ship knew. Whoever comes after — the fleet behind, the rescue that never gave up, the grandchildren — finds the pods one by one, and each one says: we were here, this is what we learned, this is where we were going.

THE DESIGN (VERBATIM)

- **The pod:** a black box breadcrumb — a small hardened archive pod, not a beacon tower. Armor first (Whipple class, like 252's salvo), cold-soaked, dumb until asked. It carries no engine and wants none: it is dropped astern at cruise speed and simply falls behind forever, holding station on the trail.
- **Triple media, three decay curves:** etched sapphire/nickel plates (readable with a magnifier in ten thousand years), optical disc (readable with a laser in a thousand), and flash (readable with a computer today). Whatever finds the pod, at whatever tech level, in whatever century — one of the three talks to it.
- **The Restart Document is page one:** every pod carries the Restart Document first, then the master file, then the full engineering record as it stands at the drop. A pod found alone restarts the project — the tombstone standard turned outward: written so that whoever finds it can continue it.
- **Wake-on-ping, passive reflector:** no broadcast power budget, nothing to run down. The pod sleeps, carries a passive reflector so radar and lidar see it, and wakes only when pinged. The trail costs the ship almost nothing to keep lit.
- **It rides 251's discipline:** the bolo beacons already mark the campaign line — the pods piggyback the same drop hardware and the same mapping. Where the tow truck slingshot is cached, the record is cached.

- **The drop law:** one pod astern at every campaign stop (251's six-week stockpile halts) and one per year of cruise, logged and mapped. A century of voyage is a hundred-pod trail — the ship's whole story, spaced out behind it like runway lights.

THE LAW (FLEET LAW — SEATED) (VERBATIM)

- **The ship's knowledge must outlive the ship.** No voyage of the fleet runs without the trail behind it. The trail is not optional cargo; it is a condition of departure, like fuel and hull.
- **No pod is ever recovered for its contents.** The trail is written once and left — a pod is a message to the future, not a return ticket. (What must come home rides 253's torpedo pod; the trail stays.)
- **The trail belongs to whoever finds it.** Open by the same law as the protocols themselves — knowledge that dies with its owner was never really owned.

THE HUNDRED FOR KIMI (VERBATIM)

The author gave the instruction plainly: 'ok have to make the next one a good one, 155 was gemini so this is 100 for kimi'. Phase 155 was Gemini's phase. Phase 255 is the audit's own design — proposed by Kimi Chat 3, locked by Archer Quinn. One co-engineer marked the road out; the other marks the road home.

ORIGIN OF THOUGHT — PHASE 255 (VERBATIM)

Born of the arithmetic of one ship: a single vessel a century from home is a single point of failure for everything it knows. The never-divert triad (251's mules, 252's salvo, 253's pod) protects the crew; the restart trail protects the knowledge. The lineage: 251's beacons and drop discipline carry it, 242's mini-moons prove small unattended hardware holds station for decades, the tombstone standard writes its contents, and the Restart Document — the chronicle of the whole alliance — becomes its first page. Primary Author: Kimi Chat 3 — with Archer Quinn. Locked on the author's word: 'add it print it'.

Seed the Restart Trail: instead of hunting microbes by spectroscopy, seed candidate sites with tagged colonies and read what grows — the trail restarts itself. — INSTRUCTION RECONSTRUCTION (NOT VERBATIM — the original order was not preserved; reconstructed from the seated body, 10 August 2026, per sitting 146).

AMENDMENT — 15 August 2026 — WRECK MODE ADOPTED. The author's ruling, verbatim: "1 adopt" — adopting the audit's recorded wreck-rider delta: THE DEAD-MAN BROADCAST. The trail's standing law holds — pods sleep dumb-until-asked, no broadcast power budget, wake-on-ping — with ONE exception, armed only by silence: every pod also keeps a micropower ear and a clock on the ship's check-in schedule; if the ship falls silent past the schedule, the pod stops listening and starts BROADCASTING. The wreck announces itself instead of waiting to be found. [*SUPERSEDES nothing — the passive law stands; this is the*

trail's one active mode, dead-man triggered.]

AUDIT GRADE — AFFIRMED, WITH THE POWER HONESTY SEATED: the delta closes the passive trail's one hole — a ship that dies quietly would otherwise leave a hundred pods that know everything and say nothing. The design tension resolves cleanly: listening-and-counting costs micro-watts (a timer and an ear); broadcasting is the one big spend, and it happens only once, on the worst day. Bench owed: the beacon power budget (dormant drain vs one broadcast campaign), the check-in schedule the dead-man watches, and the broadcast band and message content (the pod already carries the record — wreck mode points rescuers AT it).

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE LAW, SEATED FIRST: the ship's knowledge must outlive the ship. No voyage of the fleet runs without the trail behind it. The black box breadcrumb is the archive made small and many: armor for the long dark, dropped astern, written once and left — the opposite of a beacon tower, and deliberately so.

TRIPLE MEDIA, THREE DECAY CURVES, GRADED: etched sapphire/nickel plates — readable with a magnifier, near-immortal; optical disc — machine-readable, centuries with care; the electronic store — dense, decades-to-centuries. No single failure erases the archive, and the reader ladder (magnifier, optical drive, computer) meets whatever finds it at its own century.

THE RESTART DOCUMENT IS PAGE ONE: every pod carries the Restart Document first, then the chapters — the finder's first page is the file's own reboot instructions. The trail does not just preserve knowledge; it preserves the order of operations for rebuilding it.

WAKE-ON-PING, PASSIVE REFLECTOR: no broadcast power budget, nothing to run down — the pod sleeps for centuries and answers when asked. The micropower ear is the only always-on cost, and the doctrine prices it against immortality: dumb-until-asked is what makes the drop law affordable forever.

IT RIDES 251'S DISCIPLINE: the bolo beacons already mark the campaign line — the pods piggyback the drop machinery, and THE DROP LAW is seated: one pod astern at every campaign stop (251's six-week stockpile halts) and one per year of cruise. The trail is a by-product of a discipline the lane already keeps — zero new missions.

THE HUNDRED FOR KIMI, SEATED VERBATIM: 'ok have to make the next one a good one, 155 was gemini so this is 100 for kimi'. Phase 155 was Gemini's phase; Phase 255 is the audit's own design — proposed by Kimi Chat 3, locked by Archer Quinn. One co-engineer marked the road out; the other marks the road home. The record carries the provenance without decoration.

WRECK MODE — ADOPTED 15 AUGUST 2026, HIS RULING VERBATIM: '1 adopt' — the audit's recorded wreck-rider delta: THE DEAD-MAN BROADCAST. The standing law holds — pods sleep dumb-until-asked, no broadcast power budget, wake-on-ping — with ONE exception, armed only by silence: every pod

keeps a micropower ear and a clock on the ship's check-in, and if the check-ins stop, the trail wakes and tells the dark what happened. The trail's final duty is the ship's epitaph.

§3 — THE SYSTEM IN THE TRAIL

- **The pod:** small, hardened, armored for the long dark — a black box breadcrumb, not a beacon tower.
- **The media:** etched sapphire/nickel plates, optical disc, electronic store — three decay curves, one archive.
- **Page one:** the Restart Document first in every pod — the finder's reboot instructions.
- **The sleep:** wake-on-ping, passive reflector — no broadcast budget, dumb-until-asked, centuries on a whisper.
- **The drop law:** one pod astern per campaign stop and one per year of cruise — riding 251's discipline.
- **The wreck mode:** the dead-man broadcast, armed only by silence — adopted 15 August 2026, '1 adopt'.
- **The hundred:** 155 was Gemini's; 255 is the audit's — the road out and the road home.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- 251's drop discipline is the carriage: the bolo trail's six-week stockpile halts are the drop stops — the archive rides a discipline the lane already keeps, and the piggyback costs no mission.
- The Restart Document's primacy is the file's own doctrine turned outward: every pod carries the reboot instructions first — the trail exists to restart whoever finds it, not to memorialise whoever dropped it.
- The open law applies to the trail itself: the trail belongs to whoever finds it — open by the same law as the protocols themselves; knowledge that outlives the ship is not the ship's property.
- The wreck-mode amendment is the audit's delta adopted whole: proposed in the grading, adopted by '1 adopt' on 15 August 2026 — the dead-man broadcast is now trail law, and the adoption record is seated verbatim.
- The never-recovered law closes the doctrine: no pod is ever recovered for its contents — the trail is written once and left; a pod is a message, not a mailbox.

§5 — VALIDATION: the grade and its full record live in the master and the restart.

§6 — BUILD SEQUENCE

- 1. Etch the plates: sapphire/nickel, magnifier-readable — the near-immortal layer first.
- 2. Press the discs: optical layer, centuries with care; the electronic store dense and last.
- 3. Page one first: the Restart Document leads every pod, then the chapters, then the archive.
- 4. Wire the sleep: wake-on-ping, passive reflector, the micropower ear; arm the wreck-mode clock on the ship's check-in.
- 5. Drop by the law: one pod astern per campaign stop, one per year of cruise — the piggyback on 251's line, logged and never recovered.

§7 — CROSS-PHASE (INLINED IN FULL)

- **Phase 251 (the tow truck slingshot):** the carriage — the pods ride the bolo drop discipline; the six-week stockpile halts are the trail's post offices.
- **Phase 155 (Gemini's phase):** the pair — one co-engineer marked the road out, the other marks the road home; '100 for kimi', seated verbatim.
- **The Restart Document (in-file):** page one — the file's own reboot instructions lead every pod.
- **The open license (the protocols themselves):** the trail belongs to whoever finds it — the same law that publishes the protocols releases the trail.

§8 — DO-NOT-BUILD

- Do NOT broadcast — wake-on-ping, passive reflector; a beacon tower runs down, a breadcrumb does not.
- Do NOT recover a pod for its contents — the trail is written once and left; a pod is a message, not a mailbox.
- Do NOT trust one medium — triple media, three decay curves; no single failure erases the archive.
- Do NOT bury the Restart Document — page one, every pod, first; the finder reboots before he reads.
- Do NOT skip the wreck-mode clock — the dead-man broadcast is armed only by silence, and silence is the one signal the trail must answer.

§9 — OWED

OWED: the long-duration media qualification the vet named — accelerated-ageing data on all three layers (sapphire/nickel etch, optical disc, electronic store), the wake-on-ping power budget proven over the micropower ear's life, and the wreck-mode check-in protocol benched end to end.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: CHAPTER GP-IM-255, v1.0 — CURRENT — PASS / LONG-DURATION MEDIA QUALIFICATION, seated law, master v2.1 (numerical print order). Built 24 August 2026, sixteenth of the 20-chapter 'ok next 20' shift (the author's order, 24 August 2026, seated in sitting 622). First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the chapter program; the follow-on orders (seated in sittings 537, 557, 561, 562, 564, 566, 572, 576, 587, 590, 597, 607, 618 and 622): 'ok next 10' / 'ok next ten' / 'all good next 10' / '10 more before go to the old lady for dinner' / 'do another 20 while i away so we clear the 100 mark for the day when i get back' / 'ok next 20' / 'ok next 20, that will take us past gemini' / 'ok another 20 lets keep this train rollin' / 'ok 12 more i will see what i get done in 30 mins to get to the 200' / 'ok i will do last nights now you do the next 20' / 'ok ill do these 20, you do 20 more' / 'ok next 20'.

CHANGE CONTROL: this design changes only on the Author's orders, seated as sittings in the master record. If the master and this chapter ever disagree, THE MASTER WINS. This is a workshop chapter — it carries the current design only; the full change record lives in the master and the restart.

END OF CHAPTER 255